

TECHNICAL REFERENCE

Architecture Deep Dive

uPrep Platform

PROJECT	uPrep Platform
GENERATED	August 5, 2026
VERIFICATION	Verified against live source + running production deployment

uPrep · Architecture Documentation

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uPrep Platform — Architecture Deep Dive

Status	Verified against live source + running production deployment
Scope	<code>platform/web</code> (Next.js app), <code>platform/mobile/android</code> (native app), <code>platform/deploy</code> (infra), legacy <code>lms-master</code> monorepo (proxied services)
Verification method	Direct source reads, live grep/counts against the repository, live queries against the production MongoDB and legacy services over SSH. No claim below is from memory alone — see the "Evidence" note under each numbered fact.
Not verified	Legacy modules <code>billing/</code> , <code>comm/</code> , <code>social/</code> , <code>event/</code> , <code>viewer/</code> , and <code>cmds/cmds-services</code> exist in source but are not started by the current deployment and not called by the rebuild (confirmed by absence from <code>entrypoint.sh</code> and <code>lib/config.ts</code>). This document does not describe their internal behavior — only that they exist and are currently dormant.

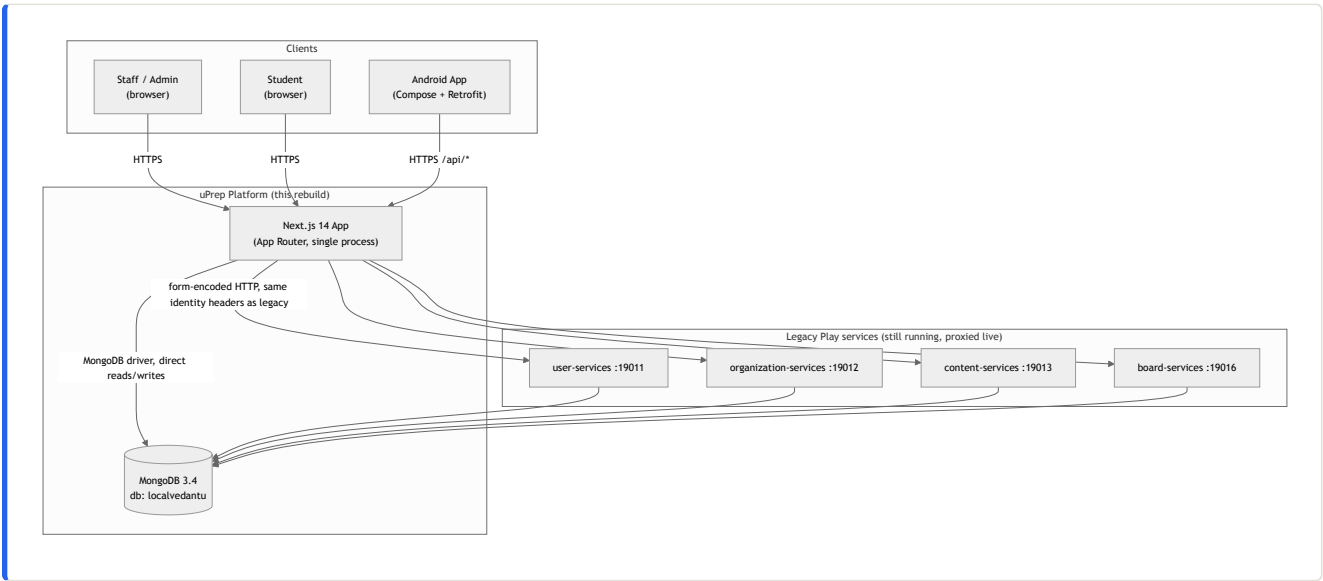
1. What this system is

uPrep is a B2B SaaS platform for test-prep coaching institutes (JEE/NEET-style exam prep). It has two audiences sharing one backend:

- Staff/Admins** — run a coaching institute's back office: enroll students, build question banks, assemble tests, manage content, view analytics. ("CMDS" = Content Management & Distribution System, the legacy product name, kept in this rebuild.)
- Students** — consume content (videos, e-books, tests), take tests, ask doubts, track their own progress.

It is a **multi-tenant** system: every document in the database is scoped to an `orgId` (one coaching institute = one org), and a super-admin role can operate across orgs (granting content between institutes, creating new institutes).

2. Context diagram



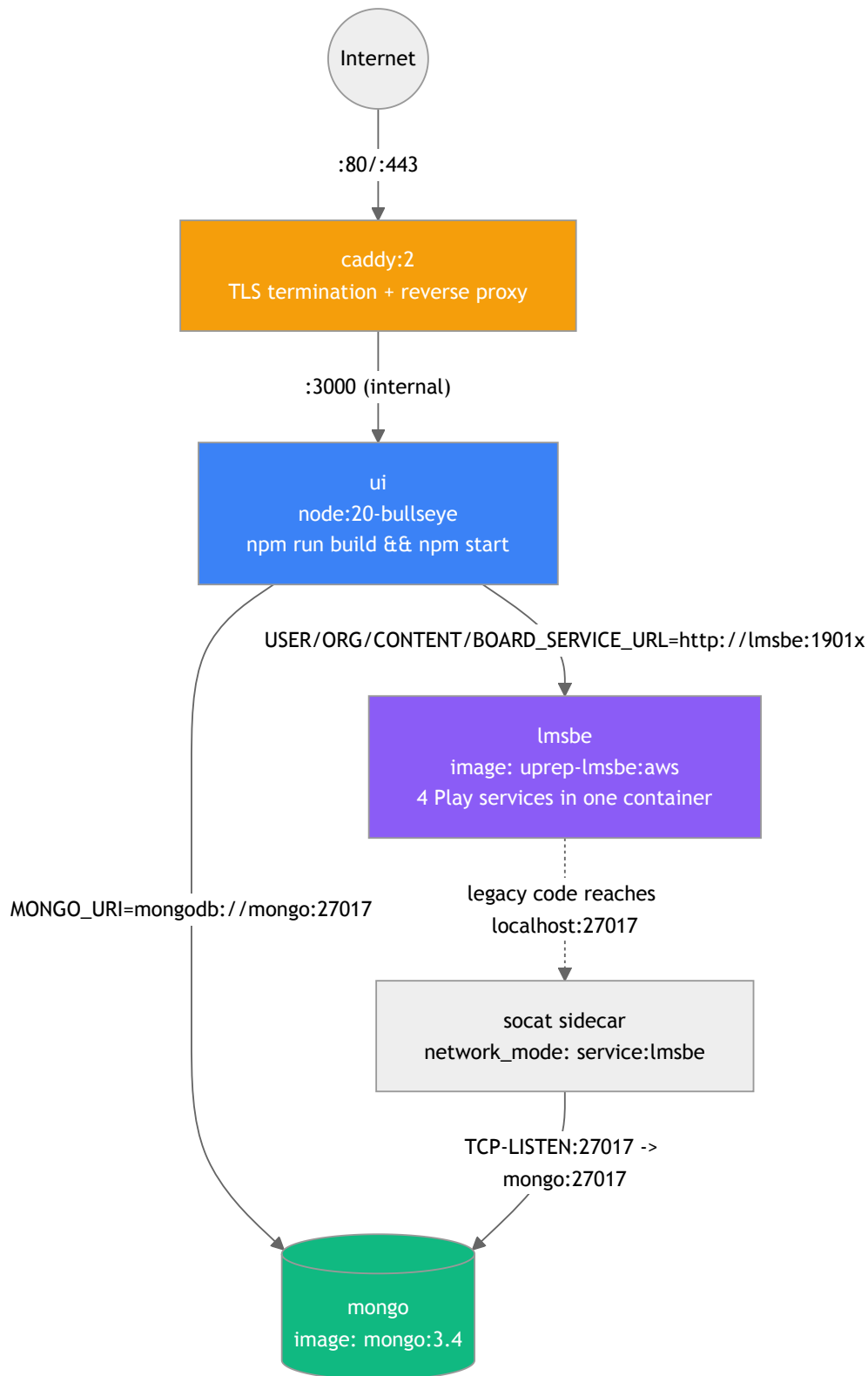
Evidence: `lib/config.ts` (the four service URLs + Mongo URI), `platform/deploy/entrypoint.sh` (starts exactly these four Play services, on exactly these ports, and no others).

3. Architectural patterns in use

These are the patterns actually implemented, not aspirational:

Pattern	Where	Why
Strangler Fig	The whole rebuild	The two legacy front-door web apps (<code>ui/cmds-app</code> , <code>ui/learn-app</code> — Play + server-rendered templates) are being replaced by the Next.js app one feature at a time, while the underlying legacy <i>data</i> services (<code>user/org/content/board</code>) keep running and are called over HTTP for the parts not yet natively reimplemented.
Backend-for-Frontend (BFF)	<code>app/api/**</code> (91 route handlers)	Next.js API routes are the only backend the browser/Android app talk to. They either read Mongo directly or fan out to legacy services, and shape the response for exactly what the UI needs — the client never talks to Mongo or the legacy services directly.
Adapter / Anti-Corruption Layer	<code>lib/legacyOrg.ts</code> , <code>lib/legacyBoard.ts</code>	Each wraps one legacy service's form-encoded, Java-DTO-shaped protocol behind a small typed TypeScript function, so the rest of the app never constructs legacy request payloads by hand.
Edge-gated RBAC	<code>middleware.ts</code>	A single middleware function gates every <code>/cmds/**</code> page and <code>/api/cmds/**</code> call on one condition (staff profile), mirroring legacy's <code>Security.checkAccess()</code> <code>@Before</code> interceptor — but running at the edge, before any page code executes.
Signed, stateless session token	<code>lib/auth-session.ts</code>	HMAC-SHA256-signed cookie carrying identity + org + role, verified with Web Crypto (works identically in the Node route runtime and the Edge middleware runtime). No server-side session store.
Direct data access, no ORM	Every route under <code>app/api/**</code>	The MongoDB driver is used directly (<code>db.collection("x").find(...)</code>) — there is no ORM/repository layer. Collections and their shapes are the closest thing to a schema this system has.
Multi-tenancy via row/tenant scoping	<code>lib/org-scope.ts</code> , every collection's <code>orgId</code> / <code>contentSrc.id</code> field	Not database-per-tenant or schema-per-tenant — every document carries its owning org, and <code>resolveOrgId()</code> pins a normal admin to their own org while allowing a super-admin to override it.

4. Deployment architecture



Evidence: `platform/deploy/docker-compose.yml` (read in full), `platform/deploy/entrypoint.sh`.

Facts worth calling out explicitly (all directly from the compose file):

1. **Single host, single docker-compose stack** — five containers (`mongo`, `lmsbe`, `socat`, `ui`, `caddy`) on one bridge network (`lmsnet`), no orchestrator, no horizontal scaling.
2. **ui builds itself on container start** — the command is literally `npm install && npm run build && npm start`, not a pre-built image. There is no CI pipeline building a versioned artifact; a deploy is "sync source,

recreate container, it rebuilds itself."

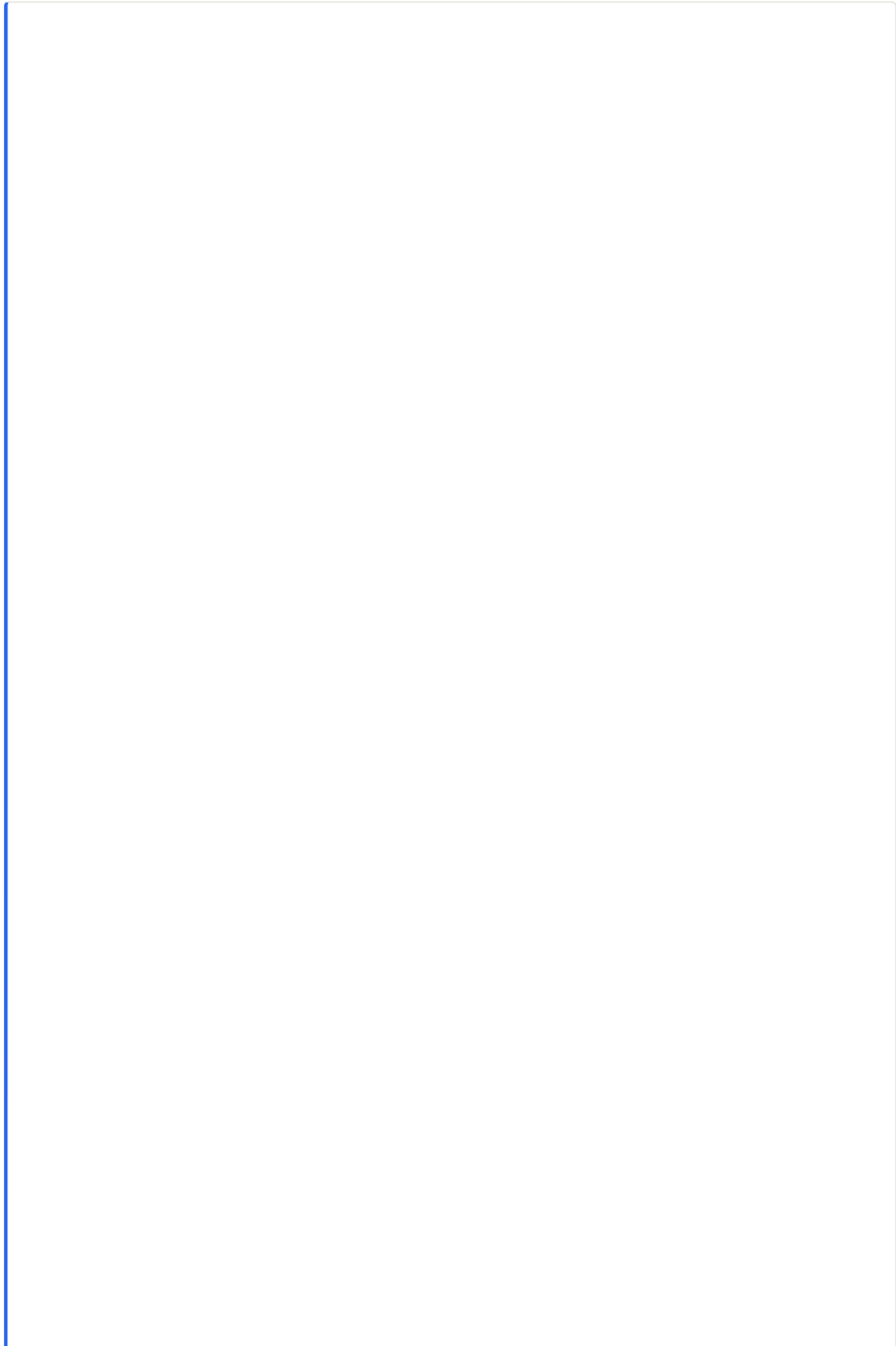
3. **lmsbe** is one container running four independent legacy JVM services (ports 19011/19012/19013/19016) via `sbt start <port>` per service, not four separate containers.
4. The **socat** sidecar exists purely because legacy `local.conf` hardcodes `localhost:27017` — it shares `lmsbe`'s network namespace and forwards `127.0.0.1:27017` to the real `mongo` service, so the legacy JVM code doesn't need to be touched to reach a container-networked Mongo.
5. **MongoDB is version 3.4** (confirmed live: `db.version()` → `3.4.24`), matching the `mongo:3.4` image pin — this is a legacy-mandated floor, not a current choice (see §12, Risks).
6. **The actual deploy mechanism** (observed and executed repeatedly in this project) is: `rsync` the `platform/web` source tree to the host, then `docker compose up -d --force-recreate ui`, which re-runs the build-and-start command above. There is no blue/green or rolling deploy — the container restarts with a brief gap.

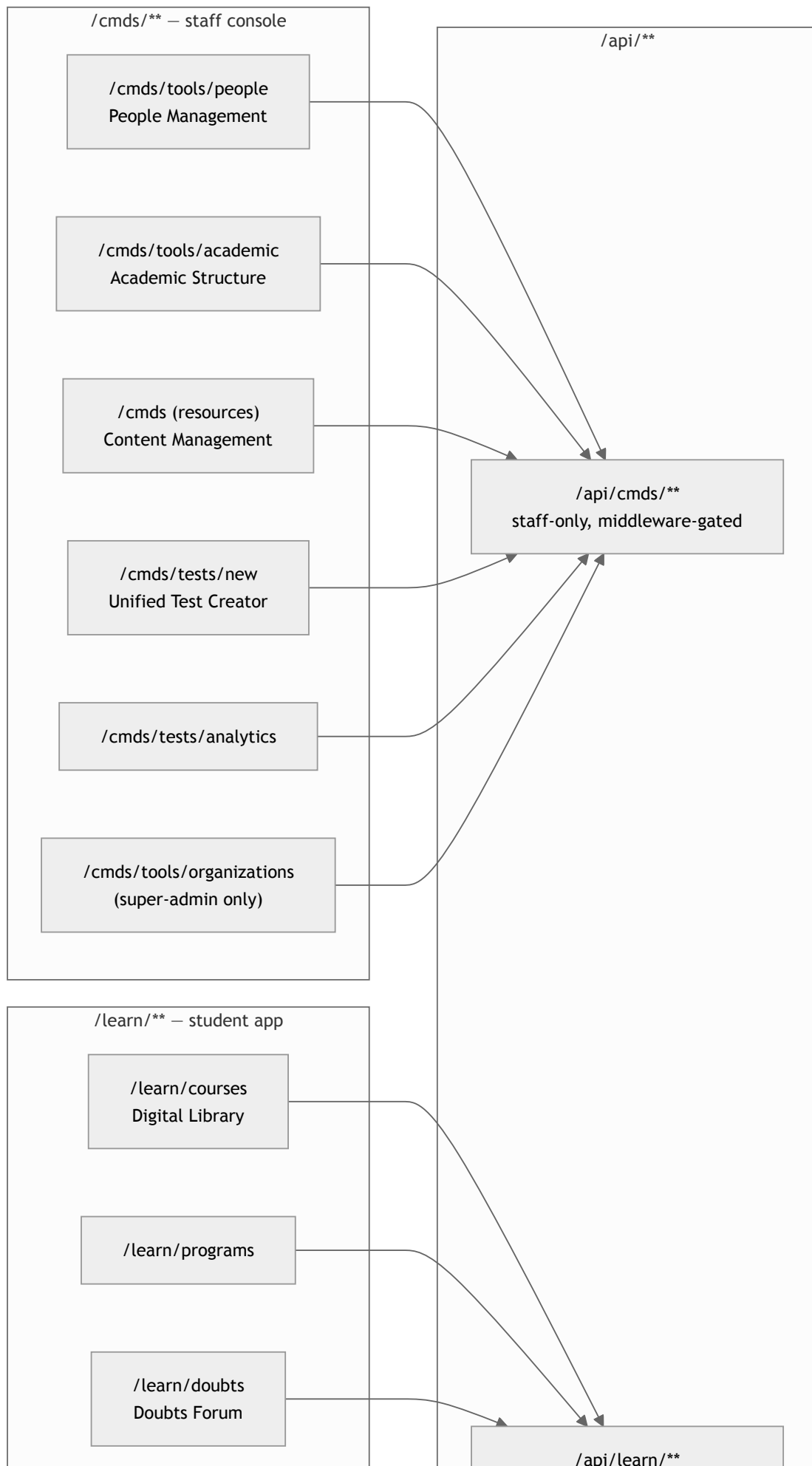
5. Application architecture (the Next.js app)

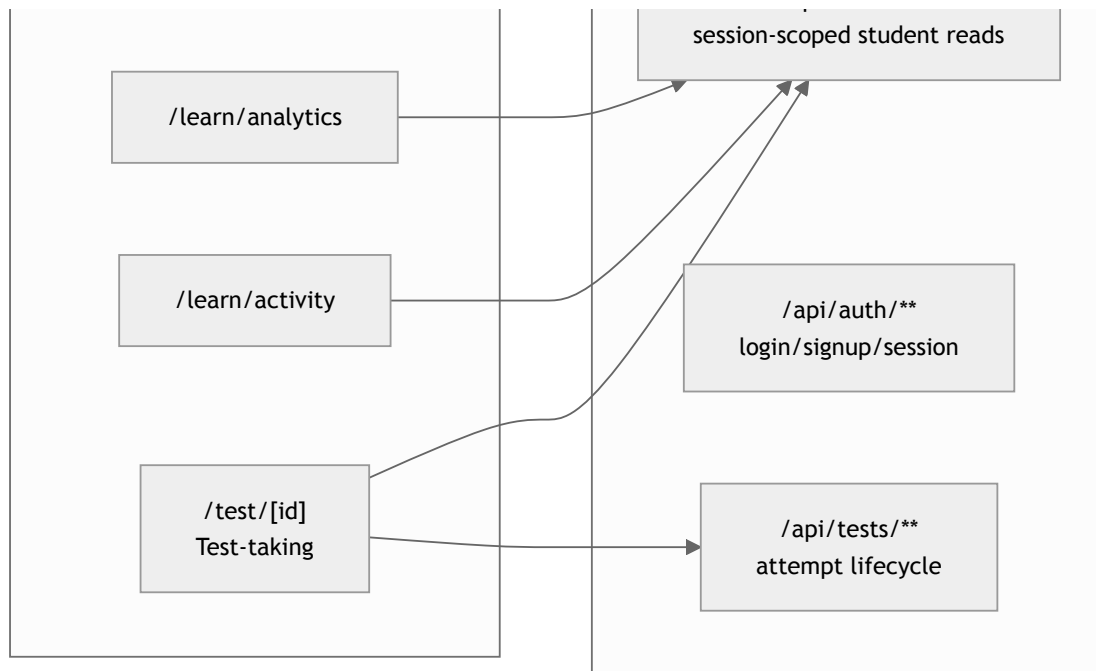
Evidence: direct `find` / `wc -l` counts against `platform/web`, `package.json`.

Metric	Count
Page routes (<code>page.tsx</code>)	74
API route handlers (<code>route.ts</code>)	91 (was 92 at audit time — see §6.2; one dead, vulnerable route was deleted rather than fixed)
— under <code>/cmds/**</code> (staff)	39 pages
— under <code>/learn/**</code> (student)	23 pages
Shared domain modules (<code>lib/*.ts</code>)	21
Shared UI components (<code>components/*.tsx</code>)	10
TypeScript/TSX lines of code	~34,200

5.1 Route-space split





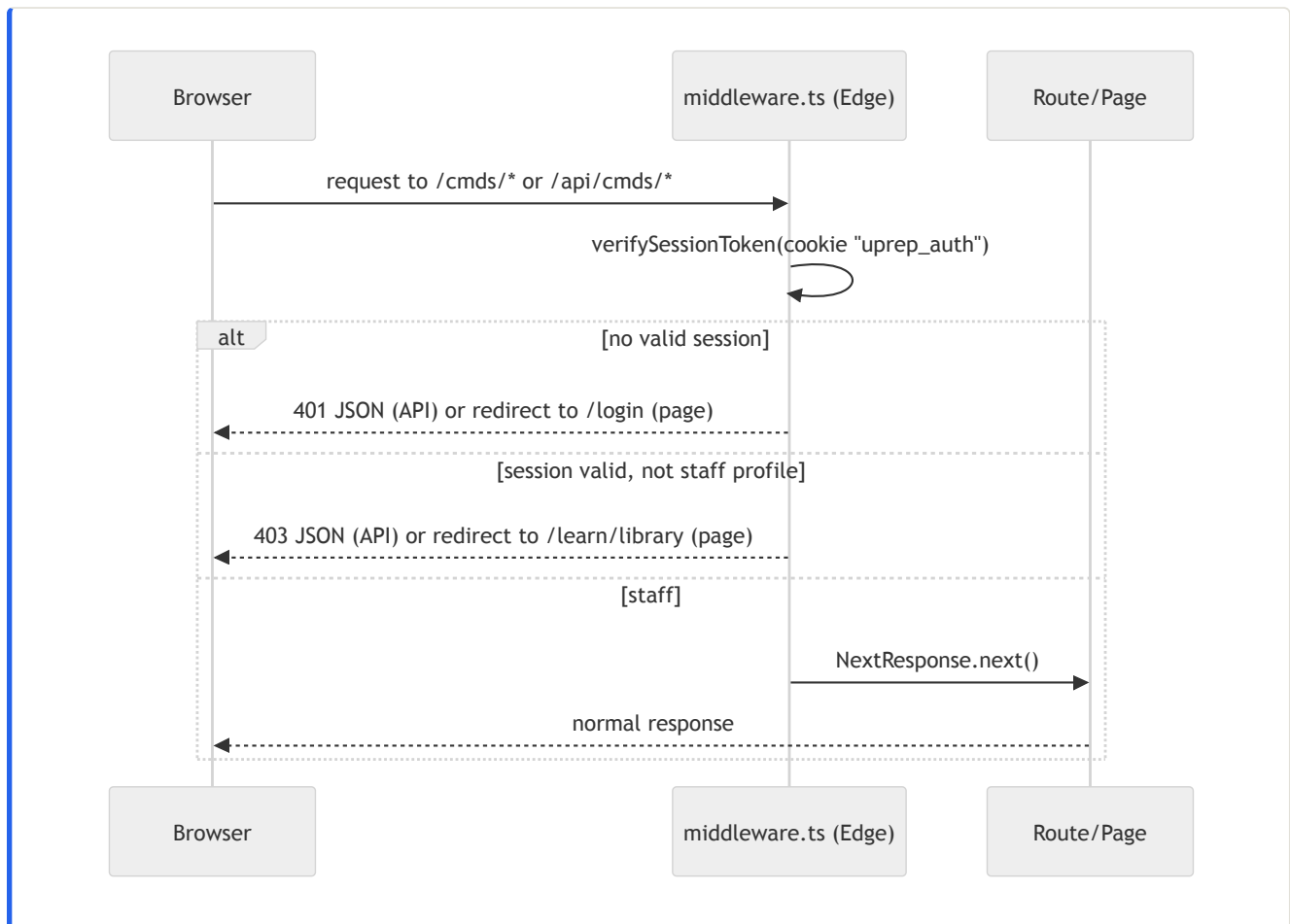


5.2 The `lib/` layer (closest thing to a service/domain layer)

There is no formal service-layer abstraction, but `lib/` consistently plays that role. The load-bearing modules, by responsibility:

- **Identity & access:** `auth-session.ts` (token sign/verify), `server-session.ts` (cookie → payload for route handlers), `session.ts` (client-side `sessionStorage` mirror, *not* trusted for authorization), `roles.ts` (`isStaff`, `isSuperAdmin`, `canManageContent`), `org-scope.ts` (`resolveOrgId`).
- **Content graph:** `courses.ts` (folder-tree traversal: `loadOrgFolders`, `collectSubtreeIds`, natural-sort helpers), `legacyBoard.ts` (Board Tree proxy + the two-hop chapter→subject resolver built this session, `resolveBoardSubjects`), `enrollment.ts` (`resolveStudentEnrollment` — the 3-way access union described in §7.3; `resolveAllProgramGroups` for staff preview).
- **Legacy adapters:** `legacyOrg.ts`, `legacyBoard.ts` — one per proxied legacy service, each owning that service's request-shape quirks (form-encoding, array-index field names, `{result, errorCode, errorMessage}` envelope).
- **Cross-cutting:** `mongo.ts` (single cached `MongoClient` / `getDb()`), `grants.ts` (cross-org content grants), `commerce.ts`, `messaging.ts`, `sections.ts`, `password.ts` (scrypt, self-contained — not delegating to legacy auth), `testCode.ts`, `subjectColors.ts` (pure presentation), `video.ts`, `storage.ts`, `login-log.ts`.

5.3 Middleware-based access gate



One explicit, deliberate exception: `GET /api/cmds/tools/news` is public (student News page reads institute announcements through the same endpoint) — `isPublicRead()` in `middleware.ts`.

Evidence: `middleware.ts` read in full.

5.4 Complete API surface reference

Evidence: every one of the 92 route handlers that existed under `app/api/**` at audit time was individually opened and read (not sampled) to produce this table, specifically to determine each route's real authorization pattern rather than assume one from its path. This exercise is what surfaced the finding written up in §6.2; one of those 92 (`/api/analytics`) was subsequently deleted as dead code during remediation, leaving 91 today (§13).

Authorization codes used below:

- **Session** — derives identity from `sessionFromReq()` (the signed cookie); rejects with 401 if absent.
- **Session+Role** — Session, plus an explicit role/ownership check (e.g. `canManageContent`, `isSuperAdmin`, org match).
- **Gate-only** — sits behind `middleware.ts`'s blanket `/api/cmds/**` staff gate with no further per-route check (any staff profile, any org the session is pinned to).
- **Public** — intentionally open, no identity required (marketing pages, webhooks with their own shared-secret, health-style lookups).
- **Fixed** — was a client-trusted-identity vulnerability at the time of the audit; remediated in this project (see §6.2) to **Session** or **Session+Role**.

`/API/CMDS/TOOLS/**` — **ORG ADMIN TOOLING**

Route	Methods	Purpose	Authorization
/academic	GET/POST/PATCH/DELETE	Departments/Programs/Centers/Sections CRUD + course assignment	Gate-only (one MANAGER-only sub- action)
/boards	GET	Board Tree proxy	Gate-only
/channels	GET/POST	Challenge Channels CRUD	Gate-only
/commerce/coupons	GET/POST/DELETE	Coupon codes CRUD	Gate-only
/commerce/invoices	GET/PATCH	Invoices list, mark paid/cancel	Gate-only
/commerce/products	GET/POST/DELETE	Sellable products CRUD	Gate-only
/devices	GET	Device/login-activity monitor	Gate-only
/exports	GET/POST	CSV export generation	Gate-only
/news	GET/POST/DELETE	Org announcements	GET is Public (exempted in middleware); POST/DELETE Gate- only
/notifications	GET/POST	Send/list notifications	Gate-only
/org-grants	GET/POST	Super-admin cross-org content grants	Session+Role (isSuperAdmin)
/organization	GET/POST	Org profile view/edit	Gate-only
/organization/logo	POST	Upload org logo	Session+Role (MANAGER)
/organizations	GET/POST/PATCH	Multi-org admin (create orgs, set plans)	Session+Role (isSuperAdmin)
/people	GET/POST/PATCH/DELETE	Member management, seat limits	GET/POST Gate-only; PATCH Session (org match, no role check); DELETE Session+Role (MANAGER)
/people/email	POST	Bulk-email a section	Session+Role (MANAGER)
/people/password	POST	Admin password reset for a member	Session+Role (org match)
/referrals	GET/POST/DELETE	Referral codes CRUD	Gate-only
/schedule	GET/POST/DELETE	Classroom Connect schedule	Gate-only
/sections	GET/POST/DELETE	Sections CRUD	Gate-only
/seller/access-codes	GET/POST/PATCH	Offline access-code inventory	GET Gate-only; POST/PATCH Session+Role (MANAGER)

Route	Methods	Purpose	Authorization
<code>/seller/groups</code>	GET/POST	Seller distribution groups	GET Gate-only; POST Session+Role (MANAGER)
<code>/signup</code>	GET/PUT	Public self-signup config editor	Gate-only

`/API/CMDs/TESTS/**` — TEST AUTHORIZING & OPERATIONS

Route	Methods	Purpose	Authorization
<code>/tests</code>	GET/POST	List gradable questions; create a test	Gate-only
<code>/tests/[id]</code>	GET/PATCH	Edit test metadata/questions	Session+Role (<code>canManageContent</code>)
<code>/tests/attempts</code>	GET/POST	Monitor/reset in-progress attempts	GET Gate-only; POST Session+Role (MANAGER)
<code>/tests/auto</code>	POST	Instant Test Generator random pick	Gate-only
<code>/tests/grading</code>	GET/POST	Manual subjective-grading queue	Gate-only (session used only for attribution)
<code>/tests/analytics</code>	GET	Per-test result analytics	Gate-only
<code>/tests/analytics/export</code>	GET	CSV export of result sheet	Gate-only
<code>/tests/schedule</code>	GET/POST/DELETE	Schedule a test to sections	Gate-only

`/API/CMDs/**` — CONTENT & ENROLLMENT (OTHER)

Route	Methods	Purpose	Authorization
<code>/assignments</code>	GET/POST	Assignment CRUD (admin side)	Gate-only
<code>/content</code>	GET/PATCH/POST/DELETE	Institute Resources browser (folders/docs/videos/tests/modules)	Session+Role (<code>canManageContent</code> ; visibility actions need MANAGER)
<code>/content/bulk</code>	POST	Bulk section/visibility/download toggle	Session+Role (MANAGER)
<code>/enroll</code>	GET/POST	Direct course enrollment for a student	GET Session (org pin); POST Fixed → Session+Role (org match)
<code>/enroll/program</code>	GET/POST/DELETE	Program+Center+Section membership	Fixed (all three) → Session+Role (org match)
<code>/modules/[id]</code>	GET	Module detail viewer	Session+Role (<code>canManageContent</code>)

Route	Methods	Purpose	Authorization
<code>/papers/[id]</code>	GET	Printable question paper incl. answer keys	Gate-only — no <code>canManageContent</code> check, unlike sibling content routes (noted as an inconsistency worth tightening, not yet fixed)
<code>/programs</code>	GET/POST	List/create Programs	Gate-only
<code>/programs/[id]</code>	GET	Program detail	Session (org-scoped via <code>resolveOrgId</code>)
<code>/programs/[id]/analytics</code>	GET	Program-level test analytics	Session (org-scoped)
<code>/programs/[id]/marksheets</code>	GET/POST	Offline marksheet upload	Session (org-scoped)
<code>/publish</code>	POST	Publish a draft question into the gradable library	Gate-only
<code>/questions</code>	POST/DELETE	Author/soft-delete a question	POST Fixed → Session+Role (<code>canManageContent</code>); DELETE already Session+Role
<code>/questions/[id]</code>	GET/PATCH	Load/edit a question	Session+Role (<code>canManageContent</code>)
<code>/questions/extract</code>	POST	PDF/DOCX text extraction utility	Gate-only
<code>/resources</code>	GET	List authored questions/tests/modules	Gate-only
<code>/upload</code>	POST	Upload a document/video file	Gate-only
<code>/videos</code>	POST	Add a video by external URL	Gate-only

`/API/LEARN/**` — STUDENT APP

Route	Methods	Purpose	Authorization
<code>/activity</code>	GET	Recent-activity feed	Fixed → Session
<code>/analytics</code>	GET	Advanced per-student analytics	Fixed → Session
<code>/assignments</code>	GET/POST	Student assignment list + submission	Session
<code>/boards</code>	GET	Board tree proxy for doubt-tagging	Public (no personal data)
<code>/bookmarks</code>	GET/POST	Bookmarks list + toggle	Fixed (both) → Session
<code>/certificates</code>	GET/POST	Eligible/issued certificates	Session, with a deliberate public-by-id "verify a certificate" lookup mode (not a bug — see §6.2)
<code>/challenges</code>	GET/POST	Time-boxed test challenges	GET Public; POST join already Session; POST create Fixed → Session

Route	Methods	Purpose	Authorization
<code>/checkout</code>	GET/POST	Storefront + purchase invoice	Session
<code>/courses</code>	GET	Enrolled courses + folder browsing	Session
<code>/doubts</code>	GET/POST	Doubts forum list/create	Fixed (both) → Session (browse itself stays public; "asked by me" + authorship now session-derived)
<code>/doubts/[id]</code>	GET/POST	One doubt + post an answer	GET Public; POST Fixed → Session
<code>/enroll-code</code>	POST	Redeem a section access code	Session
<code>/messages</code>	GET/POST	Class chat (polling)	GET Public (org broadcast only); POST Fixed → Session
<code>/modules/[id]</code>	GET	Student module viewer	Session
<code>/notifications</code>	GET	Notifications inbox	Public (org broadcasts only, no per-user data)
<code>/password</code>	POST	Change signed-in user's password	Local-account path already Session; legacy-account fallback Fixed → Session
<code>/playlists</code>	GET/POST/PATCH	Playlist list/create/mutate	GET Public; POST Fixed → Session; PATCH Fixed → Session+Role (ownership check added, previously had none at all)
<code>/profile</code>	GET/POST	View/edit own profile	Fixed (both) → Session
<code>/ratings</code>	GET/POST	Ratings/reviews	GET aggregate Public, "mine" Fixed → Session; POST Fixed → Session
<code>/schedule</code>	GET	Live-class schedule	Session
<code>/search</code>	GET	Global content search	Session
<code>/tests</code>	GET	Scheduled-tests list	Session

`/API/TESTS/**` — TEST-TAKING FLOW

Route	Methods	Purpose	Authorization
<code>/tests/[id]</code>	GET	Test info + questions, <code>alreadyAttempted</code> flag	Fixed → Session
<code>/tests/[id]/my-result</code>	GET	Read back finished-attempt score	Fixed → Session
<code>/tests/[id]/progress</code>	GET/PUT/DELETE	Save/resume/clear in-progress state	Already Session
<code>/tests/[id]/review</code>	GET	Post-submission answer review	Fixed → Session (code comment had already claimed this; the code didn't match it)
<code>/tests/[id]/submit</code>	POST	Submit + grade a finished attempt	Fixed → Session

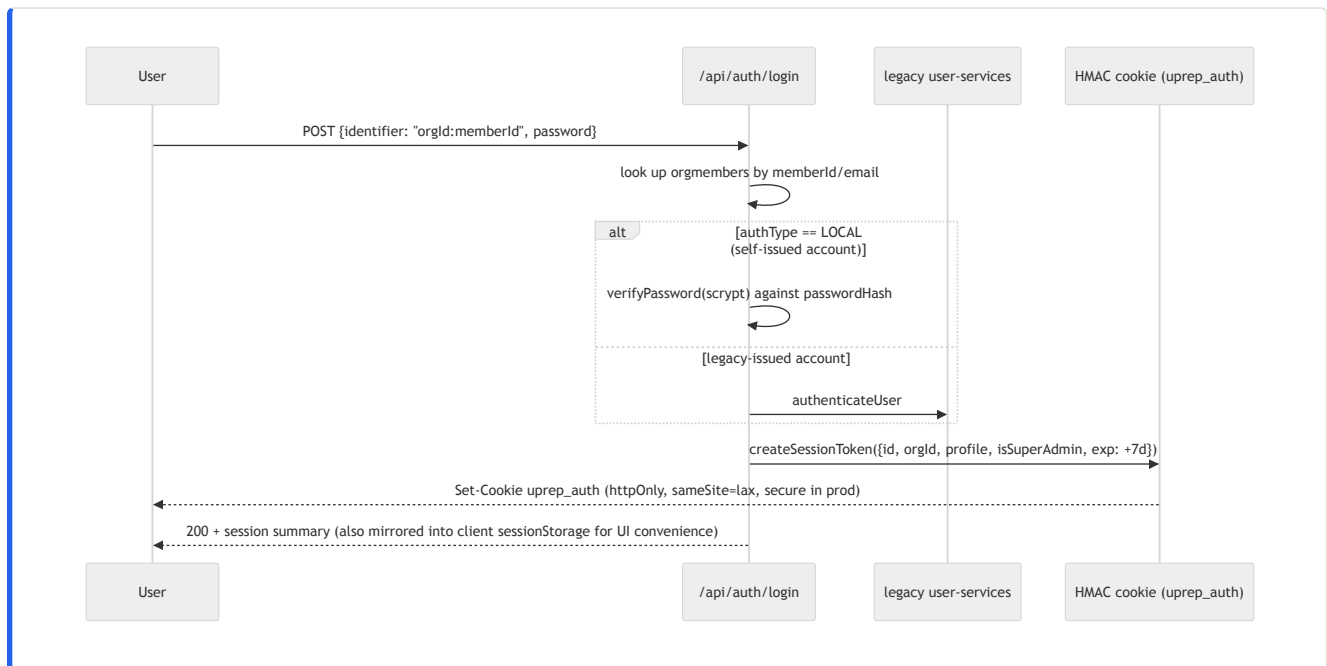
`/API/AUTH/**` — IDENTITY ESTABLISHMENT

Route	Methods	Purpose	Authorization
<code>/login</code>	POST	Password login, issues session cookie	Public (pre-auth by definition)
<code>/logout</code>	POST	Clear session cookie	Public
<code>/me</code>	GET	Return current session identity	Session
<code>/otp/request</code> , <code>/otp/verify</code>	POST	OTP login/signup	Public (pre-auth)
<code>/reset-password</code> , <code>/forgot-password</code>	POST	Password reset flow	Public, token/TTL-gated; forgot-password always responds <code>ok</code> (no account enumeration)
<code>/signup</code>	POST	Public self-registration	Public, gated by org's signup config
<code>/impersonate</code>	POST/DELETE	Staff impersonation start/stop	Session+Role (staff, org match, super-admin checks)

EVERYTHING ELSE

Route	Methods	Purpose	Authorization
<code>/api/library</code>	GET	Browsable content list, enrollment-gated	Session
<code>/api/analytics</code>	—	<i>(legacy result-history endpoint)</i>	Deleted — was vulnerable and unreferenced by any frontend code; removed rather than fixed
<code>/api/programs</code>	GET	Public program list, narrowed for logged-in students	Hybrid — intentionally public for the marketing homepage
<code>/api/seller/verify</code>	POST	Offline device/access-code verification	Public by design — identity established by the code+email+device pairing itself
<code>/api/commerce/confirm</code>	POST	Payment-gateway webhook	Shared-secret header, not a user session
<code>/api/enquiry</code> , <code>/api/orgs/suggest</code>	POST/GET	Public marketing form / org type-ahead	Public

6. Authentication & session model



Two independent password paths coexist (verified in `app/api/auth/login/route.ts` and `lib/password.ts`):

1. **Locally-created accounts** (students/teachers added via CMDs, or self-signups) — password hashed with `script`, stored as `orgmembers.passwordHash`, format `script$<saltHex>$<hashHex>`. This is a genuinely new capability the legacy stack didn't need, since legacy always owned auth.
2. **Legacy-issued accounts** — authenticated by proxying to `user-services`' `authenticateUser`.

The session token itself (`lib/auth-session.ts`) is a from-scratch addition this rebuild needed and legacy didn't: legacy ran one Play process with server-side sessions; this rebuild's middleware runs on the Edge runtime, which cannot share in-process state with the Node route runtime, so identity has to travel *in* the request as a signed, stateless token. Falls back to a **hardcoded dev secret** if `SESSION_SECRET` is unset (flagged in §12).

6.2 Authorization audit and remediation

`middleware.ts` correctly gates the `/cmds/**` surface (§5.3), but that gate only proves the caller is *some* authenticated staff member — it says nothing about *which* identity a route should act as, or whose data a `/learn/**` / `/api/tests/**` route should return. That second question is answered per-route, and a systematic read of all 92 handlers (§5.4) found it was answered incorrectly in a specific, repeated way across roughly a fifth of the API surface.

The pattern. A route would read `userId` from a client-controlled source — a `?userId=` query parameter or a `body.userId` field — and use it directly as the identity to read or write data for, instead of deriving it from the signed session cookie (`sessionFromReq()`). Because the legitimate frontend always happened to pass the caller's own id, this was invisible in normal use; nothing stopped a request from passing a *different* id.

Impact, concretely. Any authenticated (in several cases, any unauthenticated) caller could, by supplying another user's `userId`:

- Read and **overwrite** another student's profile (name, email, phone) — `GET/POST /api/learn/profile`.
- Read another student's full test-score history and per-subject weaknesses — `/api/learn/analytics`, `/api/learn/activity`, and the now-deleted `/api/analytics`.
- Read another student's **revealed answers** on a graded test — `/api/tests/[id]/review`, notably a case where the route's own comment already asserted "derived server-side from the session's userId, never trusted from the client" while the code beneath it did exactly the opposite.

- Read another student's result, or **submit and have graded a test attempt as them** — `/api/tests/[id]`, `/api/tests/[id]/my-result`, `/api/tests/[id]/submit`.
- Forge authorship of doubts, doubt answers, and chat messages under any identity — `/api/learn/doubts`, `/api/learn/doubts/[id]`, `/api/learn/messages`.
- Read/write another user's bookmarks and reviews — `/api/learn/bookmarks`, `/api/learn/ratings`.
- Mutate any playlist by id with **no authorization check of any kind**, not even an identity check — `PATCH /api/learn/playlists`.
- Inside the staff-gated area specifically: `POST /api/cmds/enroll` and all three methods on `/api/cmds/enroll/program` never checked that the *acting* staff member's org matched the *target* student's org — a valid staff session from Org A could read or mutate enrollment for a student in Org B, with only the blanket staff gate (not an org boundary) standing in the way. `POST /api/cmds/questions` authored content under a client-supplied identity with no `canManageContent` check at all, unlike every sibling content-authoring route.

Why this shape of bug, specifically. The codebase demonstrably knows the correct pattern — `/api/learn/courses`, `/api/tests/[id]/progress`, `/api/learn/search`, and several others already used `sessionFromReq()` correctly, and a few even carried comments describing a *prior* version of this exact class of bug being fixed on that route. The affected routes read as the ones a session-hardening pass missed, not a design that was never attempted.

Remediation. Every flagged route now derives identity exclusively from `sessionFromReq()`, returning `401` if no session is present; three (`/api/cmds/questions` POST, `/api/cmds/enroll` POST, `/api/cmds/enroll/program`) additionally gained the role/org-match check their siblings already had; `PATCH /api/learn/playlists` gained an ownership check where none existed before; the dead, equally-vulnerable `/api/analytics` (superseded, unreferenced) was deleted outright rather than fixed. Two routes were deliberately left as public-by-design after review rather than "fixed" — `/api/learn/certificates`'s certificate-by-id lookup (a verification feature, not a leak) and `/api/programs` (intentionally open for the marketing homepage, session-narrowed when a student is logged in).

Verification. `npm run build` (TypeScript compilation across the whole route surface) after every change; then, against the live production deployment: `GET /api/learn/profile` with no session cookie returns `401` where it previously returned another account's PII; the identical request with a valid session cookie continues to return the caller's own profile correctly, confirming no regression for legitimate use. No frontend changes were required — the browser already sends the session cookie automatically on same-origin requests, so client-supplied `userId` parameters simply became inert.

7. Data architecture

Evidence: `grep`-derived list of every `.collection("...")` call across `app/` and `lib/` (36 distinct collections found), plus live schema reads against the production database performed during this project.

7.1 Collection inventory, grouped by domain

Domain	Collections
Identity & Org	<code>orgmembers</code> , <code>organizations</code> , <code>orgprograms</code> , <code>orgcenters</code> , <code>orgsections</code> , <code>logins</code>
Content authoring (draft)	<code>cmdsquestions</code> , <code>cmdstests</code> , <code>cmdsmodules</code>

Domain	Collections
Content (published/library)	questions, tests, modules, videos, documents, questionsets, folders
Testing & assessment	userentityattempts, userquestionattempts, marksheets
Engagement / social	discussions (doubts), answers, comments, bookmarks, playlists, challenges, channels, reviews
Comms	messages, orgnotifications, news, schedules
Commerce	coupons, invoices, submissions, enquiries, referrals
Ops	exports, signupconfigs

Note: `folders` is the **content/course tree**; the **Board Tree** (subject → chapter → concept, used for tagging questions and doubts) is *not* a local collection at all — it is owned entirely by the live `board-services` legacy process and reached only over HTTP (§9.4). These are two structurally independent hierarchies that happen to share similar top-level names ("Physics XI" as both a folder and a board node with different, unrelated `_id`s).

7.2 Core entity relationships

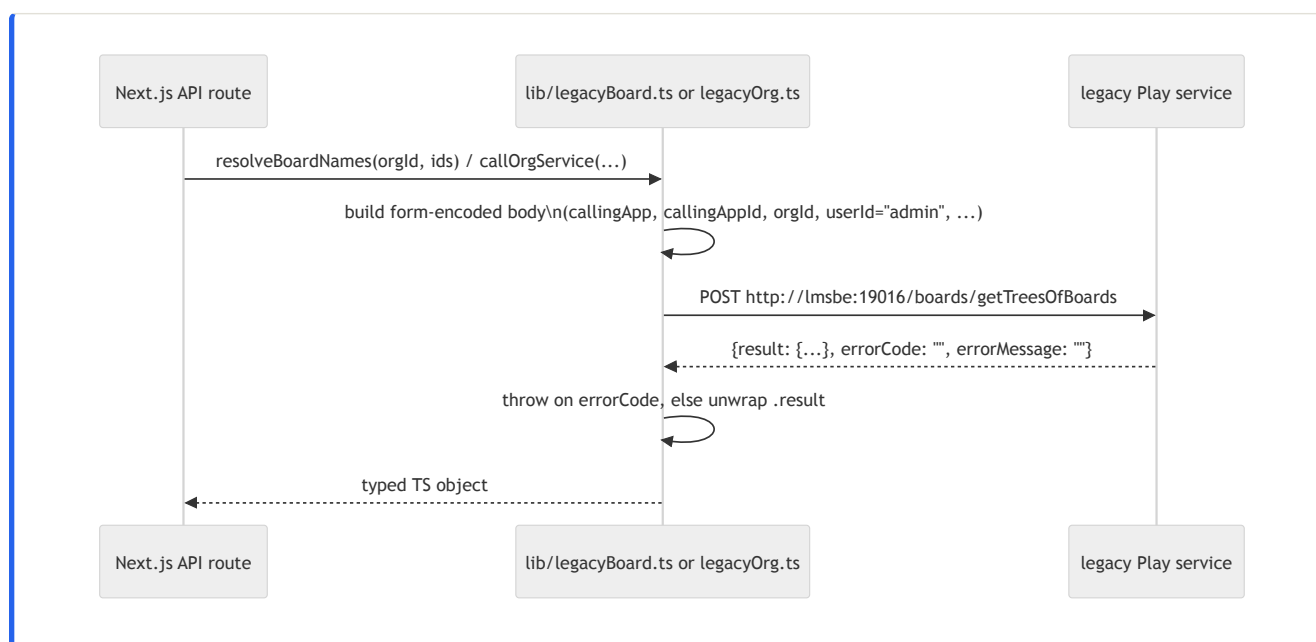
The union of these three, filtered down to the org's actual course catalog (`resolveCourseCatalog`), which also folds in cross-org grants — `lib/grants.ts`), is the student's enrolled-course root set. Every other student-facing surface (Digital Library, `/api/library` , test access checks) is required to call this same function rather than re-deriving access, specifically because two independent reimplementations of this logic drifted out of sync earlier in the project.

7.4 Test document shape

A `tests / cmdstests` document's `metadata` array is a list of **physical sections**; each section's `details` array can mix multiple question types with **independently configurable marks per type within the same section** — confirmed by inspecting a real production test document, not assumed from a naive "one section = one type" model. The JSON below is a simplified, illustrative reconstruction of that shape (field names and nesting are real; the specific values are made up for readability, not copied from a specific document):

```
{
  "name": "Weekly Physics Test",
  "metadata": [
    {
      "name": "Section 1",
      "details": [
        { "type": "SCQ", "qusCount": 10, "marks": { "1": { "positive": 4, "negative": -1 } } },
        { "type": "NUMERIC", "qusCount": 5, "marks": { "2": { "positive": 4, "negative": 0 } } }
      ]
    }
  ],
  "totalMarks": 60,
  "resultVisibility": "VISIBLE"
}
```

8. Integration architecture — calling the legacy services



Four legacy services are proxied today; each has one narrow, specific purpose in the rebuild (not "everything that service offers" — only what's been wired up):

Service	Port	What the rebuild actually calls it for
<code>board-services</code>	19016	The Board Tree (subject/chapter/concept nodes used to tag questions/doubts): <code>getChildren</code> (tree walk), <code>getTreesOfBoards</code> (batch name resolution, including the chapter→subject two-hop resolver built this session).
<code>content-services</code>	19013	Test detail + questions for the actual test-taking screen: <code>getTestInfo</code> , <code>getTestQuestions</code> (kept server-authoritative so answer keys never reach the client unfiltered).
<code>organization-services</code>	19012	Academic Structure CRUD end-to-end: <code>getOrganization</code> , <code>getDepartments</code> / <code>addDepartment</code> / <code>removeDepartment</code> , <code>getPrograms</code> / <code>addProgram</code> / <code>removeProgram</code> , <code>getCenters</code> / <code>addCenter</code> / <code>removeCenter</code> , <code>getProgramCenters</code> / <code>addProgramCenters</code> / <code>removeProgramCenters</code> , <code>getSections</code> / <code>addSection</code> / <code>removeSection</code> (<code>lib/legacyOrg.ts</code> — full action list confirmed by grep, not partial).
<code>user-services</code>	19011	Legacy-issued-account authentication fallback in the login route.

9. Key subsystems

9.1 CMDS (staff console)

People Management (with a CSV bulk-import flow that auto-generates Institute IDs and passwords), Academic Structure (Departments → Programs → Centers → Sections, plus a Courses tab for assigning content), Content Resources (question bank, tests, modules — draft in `cmds*` collections, promoted to the published collections on publish), a **unified** Create Test flow (manual pick or auto-generate share one Setup → Subjects & Types → Chapters flow, forking only on an `autoGenerateFlag`-equivalent mode toggle — this mirrors legacy's real `QrTests.createTest()` / `createTestAuto()` structure, which this rebuild originally got wrong by building two separate top-level pages before being corrected against legacy source), an Instant Test Generator (multi-subject, multi-chapter, difficulty-split auto-generation with per-question Replace), Test Analytics, and Organizations (super-admin only: create orgs, grant content across orgs).

9.2 Student app

Digital Library (subject cards grouped strictly by Program — by explicit product decision, there is no "ungrouped/Other Courses" concept), Programs, Doubts Forum, Recent Activity, and a per-student Analytics page built this session: accuracy by subject (resolved through the real board-tree hierarchy, not a hardcoded mapping) and by question type, a score trend, and derived strengths/focus-areas — none of which exist in the legacy product at all (§ see comparison document).

9.3 Test-taking engine

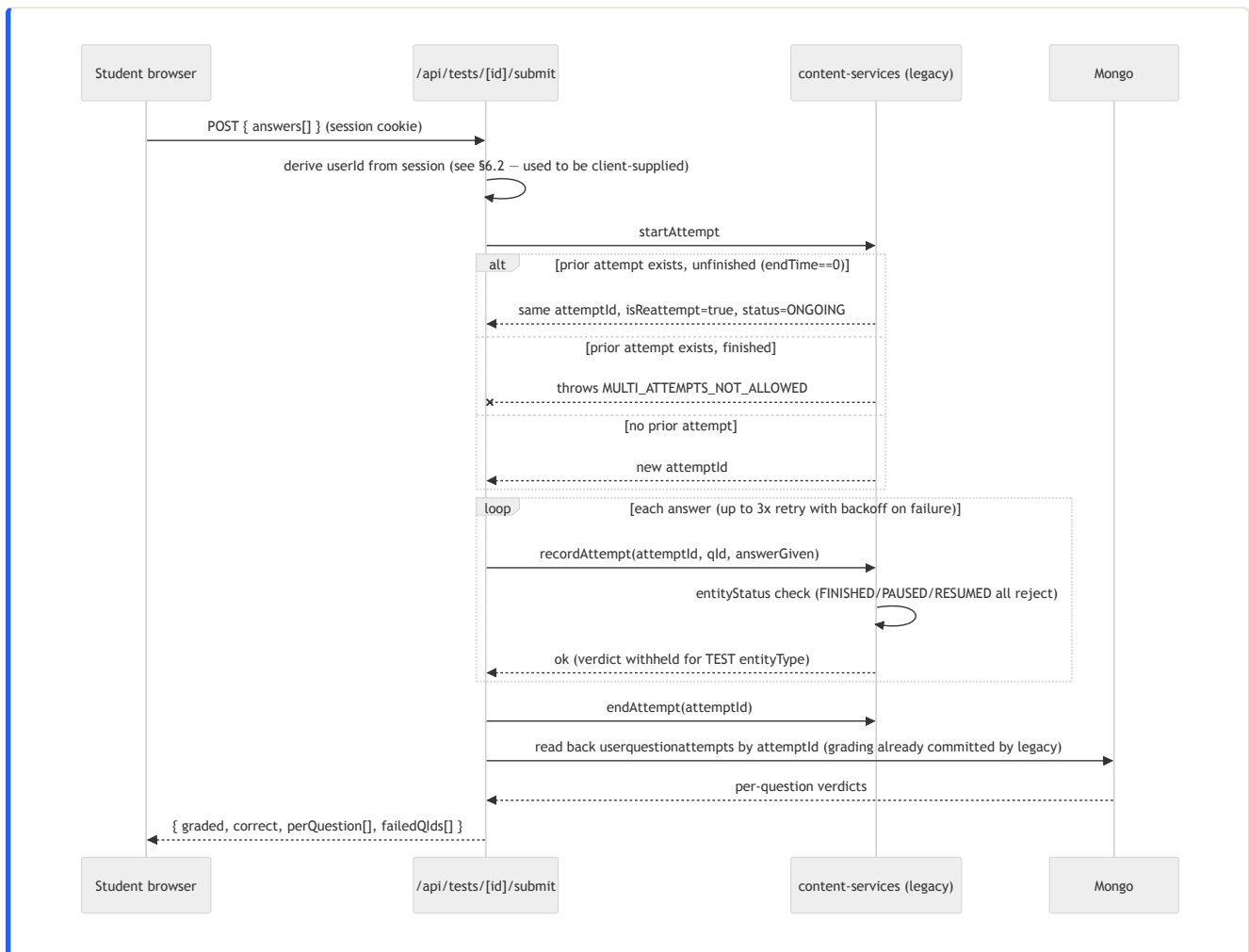
The one-attempt rule is not a rebuild policy choice — it's a **discovered legacy business rule**, and it is more precise than "one attempt, ever." Reading `AnalyticsManager.startAttempt()` (`content/content-mgmt/.../AnalyticsManager.java`, legacy) directly:

- `isMultiAttemptAllowed()` is hardcoded `return false` (line ~2820) — confirmed, not paraphrased.
- But the actual gate (`if (null != userEntityAttempt && !isMultiAttemptAllowed(...))`, ~line 302) branches on whether that prior attempt actually *finished*: if it has `endTime == 0` (abandoned/interrupted mid-test), the same attempt is silently **resumed** (`isReattempt = true`, status reset to `"ONGOING"`, same attempt id

returned) — a student who lost connection mid-test is not locked out. Only once an attempt has a real `endTime` does starting again throw `MULTI_ATTEMPTS_NOT_ALLOWED`.

- `recordAttempt()` re-checks state on *every single answer* (not just at start) — `entityStatus(attemptId)` is checked for `"FINISHED"` (throws `TEST_ENDED`), `"PAUSED"` (throws `TEST_PAUSED`), and `"RESUMED"` (throws `TEST_PAUSED_RESUME_AGAIN`) before accepting the answer. This is a defense-in-depth pattern — no single upfront gate is trusted for the whole attempt's duration.
- No MongoDB transaction wraps any of `startAttempt` / `recordAttempt` / `endAttempt` (grepped for `startTransaction` / `ClientSession` / `@Transactional` across the legacy `content/` module — no hits). Consistency across the attempt's several writes is enforced by these per-call state checks, not by atomicity.

The rebuild's own attempt path (`app/api/tests/[id]/submit/route.ts`) proxies to the same legacy `startAttempt` → `recordAttempt` (once per answer) → `endAttempt` sequence rather than reimplementing grading, and adds one thing legacy's client didn't have: each `recordAttempt` call is retried up to 3 times with backoff before being reported as failed, because a bug was found live where a silently-dropped network call left a real answer with zero corresponding `userquestionattempts` row (scored as unanswered with no error surfaced anywhere).

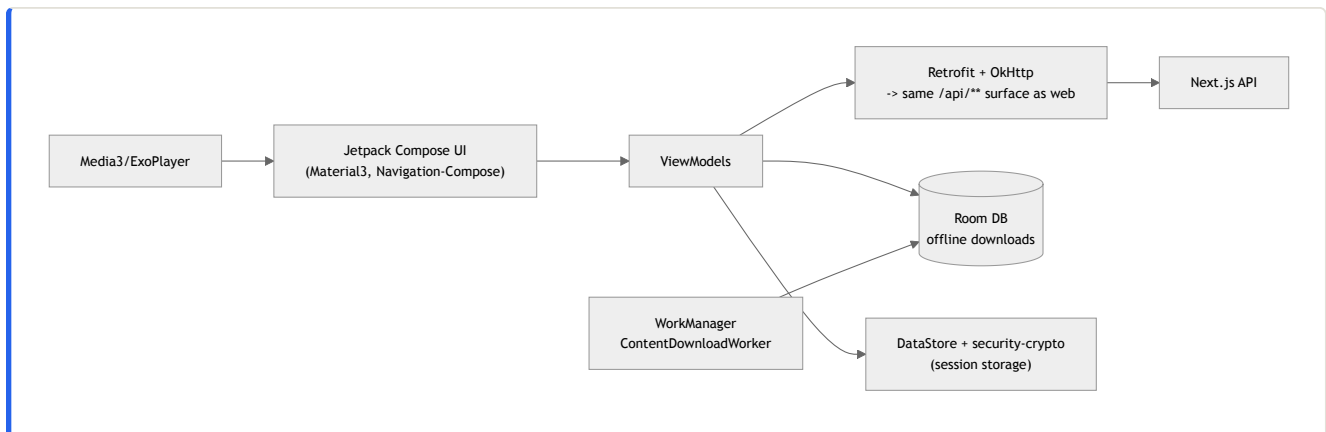


The rebuild surfaces the resume-vs-block distinction up front too (`alreadyAttempted` flag on `GET /api/tests/[id]`, computed from a direct read of `userentityattempts` rather than re-deriving legacy's logic) — an earlier, unverified version of this rebuild had no such check and let a student walk through an entire finished test only to have the final submission fail ungracefully.

9.4 Content model — two parallel trees

The **Folder tree** (`folders` collection, local) organizes *browsable content* (Program → Subject folder → Chapter folder → videos/documents/tests). The **Board Tree** (owned by `board-services`, remote) organizes *tagging metadata* for questions and doubts (Subject → Chapter → Concept). They are not the same tree, do not share IDs, and are reconciled only by a best-effort name-normalization heuristic where the UI needs to bridge them (documented in `app/cmds/tools/academic/page.tsx`'s `normSubjectName`, which the code itself notes matched 85–100% of real chapters against production data, not 100%).

10. Mobile architecture



Evidence: `app/build.gradle.kts` dependency list, file inventory (27 Kotlin files) including `DownloadDao`, `DownloadDatabase`, `DownloadEntity`, `ContentDownloadWorker`, `DownloadRepository`, `OfflineAuth`, `DownloadsScreen`, `DocumentViewerScreen` — i.e. this has moved past "WebView shell" (the README's documented original design) into native screens with offline content download, though the README itself has not been re-verified as updated to reflect this.

11. Consistency, resilience & failure modes

Evidence: direct greps against `platform/web` (`app/`, `lib/`) and the legacy `content/` module for transaction primitives, caching, retry logic, and request timeouts — reported as "not found" rather than omitted where that's the honest answer, since absence is itself the finding.

- **No multi-document transactions anywhere, in either system.** Neither the rebuild (`grep -rn "startTransaction|withTransaction|ClientSession" app lib` — zero hits) nor the legacy `content-mgmt` module (same grep pattern, zero hits) wraps a multi-write operation in a transaction. Concretely: `POST /api/cmds/publish` writes to `cmdsquestions`, then `questions`, then `answers` as three independent `insertOne` / `find` calls — if the process crashes between the second and third write, a question exists in the library with no answer key, silently ungradeable, and nothing detects or repairs that state. This mirrors the legacy attempt pipeline's own approach (§9.3): consistency is enforced by per-step state checks, not atomicity, in both systems.
- **Retry logic exists in exactly one place:** `app/api/tests/[id]/submit/route.ts`'s per-answer `recordAttempt` call (3 attempts, linear backoff), added specifically because of a live-discovered silent-drop bug (§9.3). No other route in `app/api/**` retries a failed operation.
- **No caching layer of any kind.** Grepbed for `unstable_cache`, `revalidate`, Redis, and manual memoization — none found. Every request is a fresh MongoDB round trip or a fresh legacy-service HTTP call; there is no read-through cache anywhere in the stack.

- **No explicit timeout on any legacy-service call.** None of the `fetch()` calls to `board-services` / `organization-services` / `content-services` / `user-services` (in `lib/legacyBoard.ts`, `lib/legacyOrg.ts`, or inline in route handlers) sets an `AbortController` / `signal` / `timeout`. A hung legacy service would hang the calling Next.js request indefinitely rather than failing fast.
- **Idempotency is inconsistent by design, not by oversight**, and mostly appropriately so: course-grant/program-membership writes (`/api/cmds/enroll`, `/api/cmds/enroll/program`) are upserts (safe to retry), while content-creation routes (`/api/cmds/questions` POST, `/api/learn/doubts` POST) are plain `insertOne`s with no dedupe key (a double-click or a client retry creates a duplicate row) — acceptable for admin-authored content where a duplicate is a visible, correctable annoyance, less acceptable if ever exposed to a lossy mobile network without client-side dedup.

12. Known risks / limitations (stated plainly, not softened)

1. **MongoDB 3.4.24** is years past end-of-life upstream; the version floor exists because the legacy JVM driver/queries assume 3.x wire behavior, not because it was chosen.
2. **Single VM, no redundancy** — `mongo`, `lmsbe`, and `ui` are one container each on one host; any one of them going down takes the whole platform down, and a deploy briefly stops the `ui` container.
3. **Self-building container** — the `ui` service runs `npm install && npm run build` on every start; there is no immutable, pre-tested build artifact being promoted through environments.
4. **SESSION_SECRET has a hardcoded fallback** (`lib/auth-session.ts`) if the env var isn't set — acceptable for the current single-env deployment, a real risk if ever multi-environment.
5. **No automated test suite found** in `platform/web` (verification throughout this project has been `npm run build` type-checking plus manual/live-data verification, not unit/integration tests).
6. **Runtime coupling to legacy** — four legacy Play services must be up for board-tree browsing, test-taking, and some org lookups to work at all; there is no fallback path if `lmsbe` is down.
7. **Two independent content hierarchies** (§9.4) reconciled by name-matching, not by ID — a real long-term data-integrity risk if names diverge further.
8. **No timeout on legacy-service calls** (§11) — a hung `lmsbe` process degrades to a hung Next.js request rather than a fast, visible failure.
9. **No transaction boundary around any multi-collection write** (§11) — publish, enrollment, and attempt-grading all rely on per-step state checks rather than atomicity; a mid-sequence crash can leave partial state (e.g. a question with no answer key) that nothing currently detects.
10. **The authorization audit in §6.2 was a point-in-time review, not a standing control** — nothing in the codebase (a lint rule, a route-handler wrapper, a test) currently prevents a new route from reintroducing the same client-trusted-identity pattern; the fix corrected the found instances, it didn't close off the category.

13. Appendix — verified counts

Item	Count	Source
Next.js page routes	74	<code>find app -name page.tsx wc -l</code>
Next.js API routes	91 (92 at audit time, minus 1 deleted — §6.2)	<code>find app/api -name route.ts wc -l</code>
<code>lib/</code> modules	21	<code>ls lib/*.ts</code>

Item	Count	Source
Mongo collections referenced	36	grep across <code>app/</code> , <code>lib/</code>
Web app TS/TSX LOC	~34,200	<code>wc -l</code>
Legacy Java files (whole monorepo)	2,244	<code>find legacy -name '*.java' wc -l</code>
Android Kotlin files	27	<code>find ... -name '*.kt' wc -l</code>
Legacy top-level domain modules	12 (<code>billing</code> , <code>board</code> , <code>cmds</code> , <code>comm</code> , <code>commons</code> , <code>content</code> , <code>event</code> , <code>organization</code> , <code>social</code> , <code>static-resources</code> , <code>ui</code> , <code>user</code> , <code>viewer</code>)	repo directory listing
Legacy services actually running in production	4 of those 12+ domains (<code>user</code> , <code>organization</code> , <code>content</code> , <code>board</code>)	<code>entrypoint.sh</code>